

Signaling in Education and Inequality: A Model of Social Capital Investments

13th October 2025

Abstract

The effects of income inequality on labor motivation remain debated amid stagnant social mobility. Although status is central to discussions of inequality, it is rarely endogenized in economic models, limiting understanding of its long-term interaction with inequality. This paper develops a theoretical model that combines innate ability and social capital within small-scale status competitions, where education functions as a productivity factor and a path to social status. The effects of relative income and participation costs are then explored in a dynamic hierarchy of rich and poor social positions. Results show that moderate inequality can stimulate social capital investment, but excessive inequality renders status competition unsustainable. Participation costs rise due to competition, and only two stable equilibria emerge—universal or elite-only participation. These findings clarify the tension between equality of opportunity and equality of outcome, suggesting that education policies must address signaling and participation costs to avoid reinforcing inequality.

1 Introduction

The effect of relative income on consumer behaviour has long engaged economists - but the mechanisms through which relative income or perceived inequalities shape labor participation remain contested. Some perspectives frame inequality as a necessary incentive structure that rewards effort and stimulates mobility (Baron, 2017), while others emphasize its discouraging effects on participation and long-term productivity (Gesiarz, De Neve, & Sharot, 2020). The debate is acutely reflected in education policy as well: inequality may function as an incentive that rewards effort through mobility, but it also acts as a barrier when education acts as a signaling device (Aryal, Buller, & Lange, 2021; Durst, 2021; Jaeger & Page, 1996; Hungerford & Solon, 1987).

While status motivations lie at the heart of these debates, a static treatment of status obscures long-term effects on status-driven behaviour. Individuals' investments in education and other forms of social capital respond both to perceived social position (subjective status) and to structural opportunities (objective status). In education policy, this translates into a recurring dilemma: whether to expand access by lowering participation costs or preserve excellence by rewarding high performers. So long as education carries a signaling value, status considerations cannot be ignored in assessing how inequality shapes both mobility and educational expenditures.

This paper addresses these issues by endogenising status within an economic model of social capital investment—incorporating the signaling value of education that manifests through networks, credentials, and professional affiliations. The model describes a dynamic hierarchy of positions where participants engage in probabilistic tournaments combining innate ability and acquired social capital, with success depending on both income differences and participation costs. By embedding reference-dependent utility to represent contentment with local wealth levels, the model also captures behavioural adjustments to inequality.

Three main results emerge. First, while moderate inequality may initially encourage social capital investment, excessive inequality imposes limits that make status competitions unsustainable. Second, participation costs play a central role: high costs can discourage investment even when income

differences are large, while lower costs sustain mobility-related competitions. Third, only two long-run equilibria exist—one where both rich and poor invest in status-related capital, and another where only the rich do. These findings clarify how inequality interacts with participation costs and status incentives, offering insight into the tension between equality of opportunity and equality of outcome (Heckman & Mosso, 2014) and their implications for education policy.

2 Literature Survey

The relationship between inequality, motivation, and status-seeking behavior has generated extensive debate across both economics and sociology. While early economic thought often emphasized the functional role of inequality as a motivational force—providing incentives for effort, investment, and innovation—more recent perspectives have challenged this view, particularly in the context of constrained social mobility and stratified opportunity structures. Arguments in favor of inequality (Baron, 2017) often emphasize efficiency and incentive compatibility (Becker & Tomes, 1979), based on the idea that greater rewards reflect greater effort. However, competing empirical and behavioral research suggests that beyond a certain threshold, inequality can undermine labor participation, erode social trust, and reduce long-term productivity (Perotti, 1993; Alesina & Perotti, 1996; Stiglitz, 2012). These contrasting perspectives remain visible in recent debates, reflecting an unresolved tension: whether inequality spurs economic participation or discourages it by limiting realistic pathways to advancement (Chetty, Hendren, Kline, & Saez, 2014).

Inequality has also been closely tied to education policy, often motivating efforts to expand access to schooling. The conventional argument holds that greater access fosters mobility and mitigates disparities. Yet, critics argue that if education primarily serves as a signaling mechanism rather than a productivity enhancer, such expansion may lead to wasteful expenditure rather than genuine welfare improvement (Aryal et al., 2021; Durst, 2021; Hungerford & Solon, 1987). The tension underpins a continuing debate on whether inequality stimulates effort by rewarding mobility, or suppresses it by raising barriers and entrenching stagnation. In this context, participation costs emerge as a key determinant: high costs can neutralize the benefits of reduced inequality, whereas substantial income gaps may sustain participation despite unequal opportunities (Heckman & Mosso, 2014).

Status considerations further complicate this relationship. Although status or prestige has long been recognized as a key motivator of labour participation (Bénabou & Tirole, 2003; Frank, 1993), the conventional view of status is one where it is contrasted as a social goal against the economic goals of a consumer – a dichotomy that complicates the endogenisation of status within economic models. Yet, status is neither independent of nor reducible to economic rewards (Goode, 1978; Moldovanu, Sela, & Shi, 2007). The sociological literature also emphasises this need to reconcile objective and subjective aspects of status (Wegener, 1992), highlighting how both structural position and perceived rank shape behaviour. The competition-view of status that we adopt in the paper allows us to explicitly define the scope of the effects of income inequality by considering the motivations for objective status in a hierarchy of social positions. Within this framework, status influences economic outcomes indirectly: it does not enter utility directly through wealth, but through its effect on participation and relative ranking (Postlewaite, 1998).

Perspectives in education literature also reflect these concerns – highlighting how education functions simultaneously as a productivity factor and a signal of underlying ability. Several approaches that highlight signaling or credentialist value of education contrast against the human capital theory – which views education primarily as skill acquisition. While Spence's seminal framework (Spence, 1973) has established how credentials convey information, subsequent studies have detailed signaling effects across varied institutional contexts (Durst, 2021; Heisig, 2018). The job-competition model from Thurow (1975) similarly describes how workers compete for positional advantages rather than wages, emphasizing relative position over absolute skill. Jencks (1979) has also highlighted the dual economic and social character of social capital, which mediates how inequality interacts with long-term educational investments. More recently, Carbonaro and Gamoran (2002) argue that educational success depends on both innate characteristics and social participation.

Further research by Camboni, Niu, Pai, and Vohra (2021) shows how institutional monitoring interacts with heterogeneous student types in outcomes.

The present paper builds on this literature by integrating insights from signaling and social capital into a model where status is conceptualized as a dynamic position maintained through both ability and social capital investments - subject to income inequality. By endogenizing status, the model clarifies when inequality motivates participation and when it entrenches exclusion.

3 Model

3.1 Basic Setup

We consider a game where the winner strives to maintain a rich position solely to achieve higher economic gains. The two characteristics that matter in the competition where a poorer participant can be promoted to a rich position are i) skill - subject to the social capital investment from the participant and ii) raw-talent - that is unaffected by higher expenditures. Without loss of generality, both skill s_i (enhanced with expenditure) and the raw talent r_i (which cannot be changed with higher expenditure) are assumed to be uniformly distributed in the population. With a given weight μ of raw-talent in the maximum total score considered for the win in a competition, the competitions are won with a highest score $\mu r_i + (1 - \mu)s_i$ (where r_i is raw-talent and s_i are skills that are available with higher expenditure) which has a certain probability due to distribution of r_i and s_i . We assume that the population remains constant over time (there are the same number of rich and poor in total) and that the participants are not aware of their r_i , s_i before participating in the competitions.

It is worth highlighting that in the above mechanism, a rich or a poor status by itself only has an indirect effect (through higher expenditure or budget) on the chance to be promoted (or maintaining one's rich position) - a probability that depends on the individual ability r_i and social-capital s_i . All rich participants and all poor participants have the same preferences so that the individual attributes r_i and s_i - which are not known to the participant - are not used to decide upon the expenditure (effort) towards the status game. With r_i and s_i uniformly distributed, the likelihood of any participant $i \in [1, N]$ can be formulated as the probability for their total score $\mu r_i + (1 - \mu)s_i$ being higher than any other participant - where $\mu \in (0, 1)$ is a parameter that combines social-capital s and raw-talent r so that the participant with maximum score $\mu r + (1 - \mu)s$ wins the competition. The winner of the competition acquires a rich position with economic gains so that she obtains the disposable income y_2 while the loser acquires the poor position receive $y_1 < y_2$.

To start with a simpler case, consider that in a competition with raw-talents alone, the winning participant is one who has the maximum r_i drawn from the sample N . In other words, the winner in a competition of raw-talents is a player j such that $\{r_i < r_j \forall i \in [1, N]\}$ and the probability that a given participant i of raw-skill r_i is the winner is essentially that of all other players $j \neq i$ having ability $r_{j \neq i}$ less than r_i . This is the probability

$$\int_0^1 P(r_i > r_1) \times P(r_i > r_2) \times \dots P(r_i > r_{i-1}) \times P(r_i > r_{i+1}) \dots P(r_i > r_N) dr_i$$

With a uniform distribution of skills $r \in [0, 1]$, we have $P(r_i \geq r) = r_i$ for all $i \in [1, N]$ and therefore, the probability is evaluates to $\int_0^1 (r_i)^{N-1} dr_i = \frac{1}{N}$. With more players, the chances of a particular participant winning evidently declines.

Now introducing the skills s_i , consider a threshold investment $\bar{v}$ which is necessary to be spent for the social capital to be available to a participant. While this is an arrangement that evidently favours the rich participant, there are also diminishing returns to investing $\bar{v}$ towards the richer position if naturally occurring raw talent r_i is also significant in the competition. Focusing on cases where signaling value is significant, we assume that social capital is more important than raw-talent. The foregoing analysis thus assumes that $1 - \mu > \mu$ - a condition that has been used in solutions of the model.

Assumption 1. $\mu < \frac{1}{2}$

With the threshold investment $\bar{\nu}$, the participants face different probabilities to win depending on whether they spend $\bar{\nu}$ or not. More specifically, the participant takes advantage of the skills with weight $1 - \mu$ only when she spends an amount $\bar{\nu}$. The participant might avoid the investment altogether and keep the investment towards savings instead. The behaviour of all participants depends only on ν_1, ν_2 - as they are unaware of r_i, s_i before they participate.

The result depends on the following distribution function of $x_i \equiv \mu r_i + (1 - \mu)s_i$ under the Assumption 1 where r_i and s_i are independent random variables uniformly distributed between 0 and 1 (see Appendix A).

$$F_X(x) = P(x_k < x) = P(r_k\mu + s_k(1 - \mu) < x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{x-\mu/2}{1-\mu} & \mu < x < 1 - \mu \\ 1 - \frac{(x-1)^2}{2\mu(1-\mu)} & 1 - \mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

In the game where a participant i with the maximum score x_i is the winner, the number of rich positions is fixed to $N - P$ so that only the top-scorer gets to occupy the only rich position available to be filled in every competition. The behaviour of the rich and the poor participants in the long-term can thus be explained with four different cases.

- Case I - The poor participant does not participate due to insufficient investment $\nu_1 = 0$ while the rich participant does participate by spending $\nu_2 = \bar{\nu}$.
- Case II - The rich participant does not participate with $\nu_2 = 0$ but the poor participant does participate i.e. $\nu_1 = \bar{\nu}$.
- Case III - Both poor and rich participants do not participate so that $\nu_1 = \nu_2 = 0$.
- Case IV - Both poor rich and participant participate with $\nu_1 = \nu_2 = \bar{\nu}$.

With Case I, the poor participant can win when her raw-talent (weighted with μ) is higher than the total score of all other poor participants as well as the total score of the rich participants (who compete with acquired skills). The poor are priced out of accessing any skills s in their score in Case I so that only the rich participants with investment $\nu_2 = \bar{\nu}$ have skills $s > 0$. Thus, a given poor participant must be ahead both of all of the poor participants and the rich participants so that a participant i may win only if μr_i is higher than than raw-talent μr_k in all poor participants and also exceeds the score $\mu r_k + (1 - \mu)s_k$ involving skills among the rich participants

$$P_1(\text{win}|r_i) = \prod_{k=1, k \neq i}^P P(\mu r_k < \mu r_i) \prod_{k=P+1}^N P(\mu r_k + (1 - \mu)s_k < \mu r_i) = r_i^{P-1} F_X(\mu r_i)^{N-P} \quad (1)$$

Therefore,

$$P_1(\text{win}|r_i) = \begin{cases} \frac{\mu^{N-P} r_i^{2N-P-1}}{2\mu(1-\mu)} & 0 < r_i < \mu \\ r_i^{P-1} \left(\frac{\mu r_i - \mu/2}{1-\mu} \right)^{N-P} & \mu < r_i < 1 - \mu \\ r_i^{P-1} \left(1 - \frac{(\mu r_i - 1)^2}{2\mu(1-\mu)} \right)^{N-P} & 1 - \mu < r_i < 1 \\ 0 & \text{otherwise} \end{cases}$$

With the uniform density for r_i (i.e. $f(r_i) = 1$), total probability to win can be calculated as $\int_{r_i=0}^{r_i=1} P(\text{win}|r_i) dr_i$. Since the participants do not know the values of r_i, s_i before they participate in such competitions, the total probability can be evaluated with the assumption that both r_i and s_i

range from 0 to 1. Notice also that since only one participant can win the skills-game, the probability $\int_0^1 \int_0^1 P_2(\text{win}|r_i, s_i) ds_i dr_i$ for a rich participant to win is simply $\frac{1}{N-P} - \frac{P}{N-P} \int_0^1 \int_0^1 P_1(\text{win}|r_i, s_i) ds_i dr_i$.

In Case II, when the rich person has no access to social capital (i.e. all rich participants have $\nu_2 = 0$), the poor participant has the skills-advantage in the game so the probability to win for the poor participant i is

$$\begin{aligned} P(\text{win}|r_i, s_i) &= \prod_{k=1, k \neq i}^P P(\mu r_k + (1-\mu)s_k < \mu r_i + (1-\mu)s_i) \prod_{k=P+1}^N P(\mu r_k < \mu r_i + (1-\mu)s_i) \\ &= F_X(x_i)^{P-1} \prod_{k=P+1}^N P(\mu r_k < \mu r_i + (1-\mu)s_i) \quad (2) \end{aligned}$$

In Case III, when neither the rich nor the poor participate in the game, then the game depends only on raw-talents and all participants have the same probability $\frac{1}{N}$ to win the game. This is because $\int_{r_i=0}^{r_i=1} P(\text{win}|r_i) dr_i = \frac{r_i^N}{N} \Big|_0^1 = \frac{1}{N}$ where $P(\text{win}|r_i) = \prod_{k=1, k \neq i}^P P(r_k < r_i) \prod_{k=P+1}^N P(r_k < r_i) = r_i^{N-1}$.

In Case IV, when both rich and poor participants set $\nu_1 = \nu_2 = \bar{\nu}$, then the poor participant can win only if her score x_i is above all other participants - rich or poor.

$$\begin{aligned} P(\text{win}|r_i) &= \prod_{k=1, k \neq i}^P P(\mu r_k + (1-\mu)s_k < \mu r_i + (1-\mu)s_i) \prod_{k=P+1}^N P(\mu r_k + (1-\mu)s_k < \mu r_i + (1-\mu)s_i) \\ &= F_X(x_i)^{N-1} \quad (3) \end{aligned}$$

Using symmetry arguments, this winning probability - which is the same for all participants - also turns out to be $1/N$ for all the participants.

With the rich and poor participants having disparate incomes y_1 and y_2 (so that $y_1 < y_2$), the above setup evidently puts the poor participants at a disadvantage since they may not afford to pay the costs to enhance their skills (i.e. realise the true worth of their skills s_i due to high participation costs). However, since both raw-talent r_i and social-capital s_i (the latter enhanced by costly institutional support) are randomly distributed in the population, the competition may never rule out the chance for a poor participant to win. For the same reason, the returns from investments to enhance one's skills may also be diminishing since one cannot beat raw-talent and non-enhanced skills after a certain point.

3.2 Equilibrium Conditions

The solution follows by setting total number of participants $N = 2$ and the number of poor participants $P = 1$ in Equations 1, 2 and 3 above. As before, a participant takes advantage of the skills with weight $1 - \mu$ only by spending an amount $\bar{\nu}$ and can decide to avoid the expenditure altogether in order to invest it towards the savings instead. It is easy to show that these results do not change for $P > \frac{N}{2}$. We first calculate the probability to win in Cases I, II, III and IV and then discuss the equilibrium conditions using an intertemporal additive utility function.

3.2.1 Winning Probability

Case I

In Case I, the poor participant can win when her raw-talent r_1 is higher than of the total score of rich participant (who competes with acquired skills). The poor participant is priced out of accessing any

skills s in her score so that the skills $s > 0$ is only true for the rich participant having $\nu_2 = \bar{\nu}$. Thus, the poor participant with raw-talent r_1 can win with the following probability

$$P(\text{win}|r_1) = P(\mu r_2 + (1 - \mu)s_2 < \mu r_1) = F_X(\mu r_1)$$

Notice that $\mu r_1 < \mu < 1 - \mu < 1$ is implied due to $\mu < \frac{1}{2}$ (and $r_1 \in (0, 1)$). The probability that the total score $\mu r_2 + (1 - \mu)s_2$ does not exceed μr_1 implies that only the first part of the piecewise function for $F_X(x)$ needs to be considered. Therefore, using $P(\text{win}|r_1) = \frac{\mu^2 r_1^2}{2\mu(1-\mu)}$, we evaluate $P_1(\text{win})$ as

$$P_1(\text{win}) = \int_0^1 P(\text{win}|r_1) dr_1 = \frac{\mu}{6(1-\mu)}$$

Since only one of the participants can win, the probability $P_2(\text{win})$ for the rich participant to win is simply $1 - P_1(\text{win})$. The following thus applies to all other cases (i.e. Cases II, III and IV).

$$P_2(\text{win}) = 1 - P_1(\text{win}) \quad (4)$$

Case II

In Case II, the rich person does not participate ($\nu_2 = 0$) and both the poor participants have the skills-advantage in the game. A poor participant wins by having the total score higher than the raw-talent of the rich participant (who does not acquire skills with investment). Representing $x_i \equiv \mu r_i + (1 - \mu)s_i$, we have the following probability for the poor participant to win

$$P(\text{win}|r_1, s_1) = P(\mu r_2 < \mu r_1 + (1 - \mu)s_1)$$

Notice that $P(\mu r_2 < \mu r_1 + (1 - \mu)s_1)$ (i.e. $P(r_2 < r_1 + (\frac{1}{\mu} - 1)s_1)$) is 1 after $r_1 + (\frac{1}{\mu} - 1)s_1 > 1$ (or $\mu r_1 + (1 - \mu)s_1 > \mu$) and $\frac{1}{\mu}$ in the interval $(0, \mu)$ (this is because $r_2 \in (0, 1)$). In other words,

$$P(r_2 < r_1 + (1/\mu - 1)s_1) = \begin{cases} \frac{x_1}{\mu} & 0 < x_1 < \mu \\ 1 & \text{otherwise} \end{cases}$$

The probability to win for the poor participant can be written as (see Appendix A)

$$P(\text{win}|r_1, s_1) = \begin{cases} \frac{x_1}{\mu} & 0 < x < \mu \\ 1 & \mu < x_1 < 1 - \mu \\ 1 & 1 - \mu < x_1 < 1 \\ 0 & \text{otherwise} \end{cases}$$

Using double-integral properties (see Appendix B), the total probability to win $P_1(\text{win})$ for a poor participant is evaluated as follows

$$\begin{aligned}
P_1(\text{win}) &= \int_{s_1=0}^{s_1=1} \int_{r_1=0}^{r_1=1} P(\text{win}|r_1, s_1) ds_1 dr_1 = \int_{r_1=0}^{r_1=1} \int_{s_1=0}^{s_1=\frac{\mu(1-r_1)}{1-\mu}} \frac{(\mu r_1 + (1-\mu)s_1)}{\mu} ds_1 dr_1 \\
&\quad + \int_{r_1=0}^{r_1=1} \int_{s_1=\frac{\mu(1-r_1)}{1-\mu}}^{s_1=1-\frac{\mu r_1}{1-\mu}} 1 ds_1 dr_1 \\
&\quad + \int_{r_1=0}^{r_1=1} \int_{s_1=1-\frac{\mu r_1}{1-\mu}}^{s_1=1} 1 ds_1 dr_1 \\
&= \frac{\mu}{3(1-\mu)} + \frac{1-2\mu}{1-\mu} + \frac{\mu}{2(1-\mu)} \\
&= \frac{2\mu}{6(1-\mu)} + \frac{6(1-2\mu)}{6(1-\mu)} + \frac{3\mu}{6(1-\mu)} = \frac{6-7\mu}{6(1-\mu)}
\end{aligned}$$

The probability $P_2(\text{win})$ for the rich participant to win follows from Equation 4.

Case III

In Case III, when neither the rich nor the poor participate in the game, then the game depends only on raw-talents and both participants have the same probability $\frac{1}{2}$ to win the game. This is because $\int_{r_i=0}^{r_i=1} P(\text{win}|r_i) dr_i = \frac{r_i^2}{2} \Big|_0^1 = \frac{1}{2}$. Both the poor and rich participant have the same total probability $\frac{1}{2}$ to win.

Case IV

In Case IV, when both rich and poor participants set $\nu_1 = \nu_2 = \bar{\nu}$, then the poor participant can win only if her score x_i is above that of the rich participant. Therefore,

$$\begin{aligned}
P(\text{win}|r_1, s_1) &= P(\mu r_2 + (1-\mu)s_2 < \mu r_1 + (1-\mu)s_1) \\
&= F_X(x_1)
\end{aligned}$$

$$P(\text{win}|r_1, s_1) = \begin{cases} \frac{x_1^2}{2\mu(1-\mu)} & 0 < x_1 < \mu \\ \frac{x_1 - \mu/2}{1-\mu} & \mu < x_1 < 1-\mu \\ 1 - \frac{(x_1-1)^2}{2\mu(1-\mu)} & 1-\mu < x_1 < 1 \\ 0 & \text{otherwise} \end{cases}$$

While this integral can be evaluated based on symmetry arguments (since x_i is also randomly distributed), the integral $\int_{s_1=0}^{s_1=1} \int_{r_1=0}^{r_1=1} P(\text{win}|r_1, s_1) dr_1 ds_1$ also evaluates to $\frac{1}{2}$ (see Appendix B)

$$\begin{aligned}
\int_{s=0}^{s=1} \int_{r=0}^{r=1} F_X(x_1) ds_1 dr_1 &= \int_{r_1=0}^{r_1=1} \int_{s_1=0}^{s_1=\frac{\mu(1-r_1)}{1-\mu}} \frac{(\mu r_1 + (1-\mu)s_1)^2}{2\mu(1-\mu)} ds_1 dr_1 \\
&\quad + \int_{r_1=0}^{r_1=1} \int_{s_1=\frac{\mu(1-r_1)}{1-\mu}}^{s_1=1-\frac{\mu r_1}{1-\mu}} \left(\frac{\mu r_1 + (1-\mu)s_1 - \mu/2}{1-\mu} \right) ds_1 dr_1 \\
&\quad + \int_{r_1=0}^{r_1=1} \int_{s_1=1-\frac{\mu r_1}{1-\mu}}^{s_1=1} \left(1 - \frac{(\mu r_1 + (1-\mu)s_1 - 1)^2}{2\mu(1-\mu)} \right) ds_1 dr_1 \\
&= \frac{\mu^2}{8(1-\mu)^2} + \frac{1-2\mu}{2(1-\mu)} + \frac{(4-5\mu)\mu}{8(1-\mu)^2} = \frac{4(1-\mu)^2}{8(1-\mu)^2} = \frac{1}{2}
\end{aligned}$$

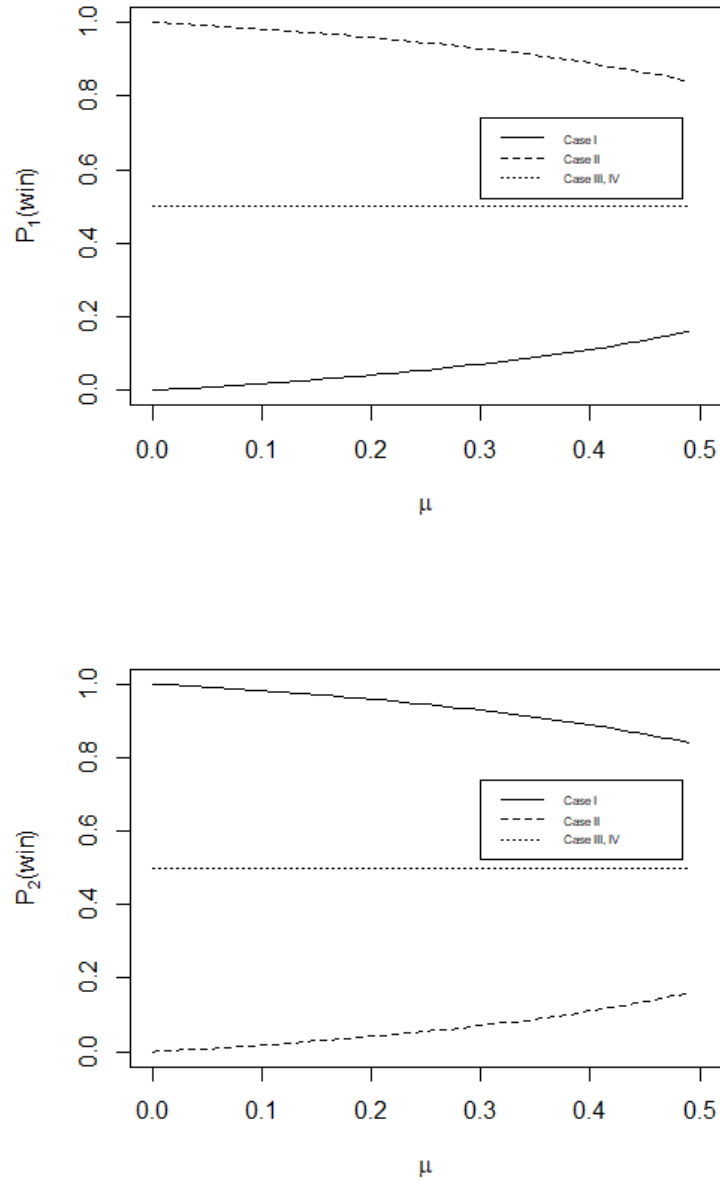


Figure 1: Probability of winning $P_1(\text{win})$, $P_2(\text{win})$ for poor and rich participants with varying μ

3.2.2 Intertemporal Choice

The participants optimise an intertemporally additive utility over their life-time - which is represented with two time-horizons. The second-time horizon is set to be one before the end of life when all wealth accumulated by either participant must disappear. With no accumulations at the end of life, both rich and poor participants must decide to spend all their incomes y_1 or y_2 (for poor and rich participants respectively so that $y_1 < y_2$) towards the promotion game. In the first-stage, however, the participants decide the amount to be spent towards promotion based upon the utility from wealth they obtain across the two time-horizons that span their lifetime. For the poor participant, the total utility to be optimised (assuming EUT) under the intertemporally additive utility $u(\cdot)$, a winning probability function $W(x)$ (for expenditure x) and a discount-factor δ is

$$u(y_1 - \nu_1) + \frac{1}{\delta}((1 - W(y_1))u(y_1) + W(y_1)u(y_2))$$

Similarly, for the rich participant, the total utility to be optimised (assuming EUT) is

$$u(y_2 - \nu_2) + \frac{1}{\delta}((1 - W(y_2))u(y_1) + W(y_2)u(y_2))$$

		Poor <i>withdraws</i>	participant <i>participates</i>
Rich Participant	<i>withdraws</i>	Case III	Case II
	<i>participates</i>	Case I	Case IV

Table 1: Cases I,II, III, IV as participant strategies

		Poor	participant
		<i>withdraws</i>	<i>participates</i>
Rich participant	<i>withdraws</i>	$u(y_2) + \frac{(1 - \frac{P}{N})u(y_2) + \frac{P}{N}u(y_1)}{\delta}$	$u(y_2) + \frac{u(y_2)p'_2 + (1 - p'_2)u(y_1)}{\delta}$
		$u(y_1) + \frac{(1 - \frac{P}{N})u(y_2) + \frac{P}{N}u(y_1)}{\delta}$	$u(y_1 - \bar{\nu}) + \frac{u(y_2)p'_1 + (1 - p'_1)u(y_1)}{\delta}$
<i>participates</i>		$u(y_2 - \bar{\nu}) + \frac{p_2u(y_2) + (1 - p_2)u(y_1)}{\delta}$	$u(y_2 - \bar{\nu}) + \frac{(1 - \frac{P}{N})u(y_2) + \frac{P}{N}u(y_1)}{\delta}$
		$u(y_1) + \frac{p_1u(y_2) + (1 - p_1)u(y_1)}{\delta}$	$u(y_1 - \bar{\nu}) + \frac{(1 - \frac{P}{N})u(y_2) + \frac{P}{N}u(y_1)}{\delta}$

Table 2: Utility for poor and rich participants

With above utilities, we now discuss the conditions for whether the Case I, II, III, IV can become Nash equilibria (see Table 1) or not - and detail the implications for the effect of income differences on social capital expenditure¹. For ease of notation we define p_1, p_2, p'_1 and p'_2 as the following:

$$\begin{aligned} p_1 &\equiv \frac{\mu}{6(1-\mu)} \\ p_2 &\equiv 1 - p_1 \\ p'_1 &\equiv \frac{6-7\mu}{6(1-\mu)} \\ p'_2 &\equiv 1 - p'_1 \end{aligned} \tag{5}$$

The choices for the poor and rich participants using a general intertemporally additive function $u(A)$ (for assets A) are shown in Table 2. As we can verify from Figure 1 that we have $p_1 < 1/2 < p_2$ and $p'_1 > 1/2 > p'_2$ for all values of $\mu < 1/2$.

Given $\mu < 1/2$, we have (6)

$$\begin{aligned} p_1 &< 1/2 < p_2 \\ p'_1 &> 1/2 > p'_2 \end{aligned}$$

Table 2 represents the condition when the budget is non-binding i.e. the poor participant is able to spend $\bar{v}$. i.e. $\bar{v} < y_1 < y_2$. When the poor participant faces a binding constraint i.e. $y_1 < \bar{v} < y_2$, then the rich participant knows that the best case probability to win for the poor participant is $\frac{1}{2}$. The rich participant's decision to participate would depend only on utility from savings and the relative advantage from gain in probability. This is elaborated in Section 3.2.3.

3.2.3 Binding constraint

Assuming EUT participants, we now consider the simpler case of binding constraint $y_1 < \bar{v} < y_2$. Considering pure strategies for participation², the rich participant would thus participate only when the utility from participating would be higher than from not participating i.e. $u(y_2 - \bar{v}) + \frac{p_2 u(y_2) + (1-p_2)u(y_1)}{\delta} > u(y_2) + \frac{u(y_2) + u(y_1)}{2\delta}$. This results in the following condition

$$\frac{(p_2 - 1/2)(u(y_2) - u(y_1))}{\delta} > u(y_2) - u(y_2 - \bar{v})$$

With income differences held constant, an increase in participation cost $\bar{v}$ discourages participation since RHS increases with $\bar{v}$ ($\because u(y_2 - \bar{v})$ is decreasing in $\bar{v}$) and may fail the above condition. Similarly, with participation cost $\bar{v}$ held constant, an increase in income-difference $y_2 - y_1$ is in the rich participant's interest whereas a narrowing of income differences reduces incentive for the participant to participate by possibly failing the above condition. All else being the same, higher impatience (lower time-discounted value of money or a higher δ) would also discourage participation. Therefore, when budget is binding for the poor participant i.e. if the poor participant cannot participate due to $y_1 < \bar{v}$, then narrower income differences, higher participation costs and higher patience for savings (common for both rich and poor participants) all encourage the rich participant to withdraw from the game. This also means that wider income differences ($y_2 - y_1$) combined with lower participation costs ($\bar{v}$) may together help maintain the equilibrium where only the rich participates. Similarly, wider income differences (higher $y_2 - y_1$) and more patience for savings (higher δ) would together maintain the incentive for the rich participant to participate.

¹The discussion of the effect of the threshold $\bar{v}$ and the income differences $y_2 - y_1$ on participation (expenditure) assumes that the exogenous parameter μ is stable with respect to short-term changes in threshold investment $\bar{v}$ as well as disposable incomes y_1, y_2 .

²With the assumption of a stability in long-term participant behaviour (which itself is a result of observed probability of success), mixed-strategies are not relevant in the current set up.

3.2.4 Non-Binding constraint

When budget is non-binding, then participation of the rich participant is her dominant strategy if the condition is also $u(y_2) + \frac{u(y_2)p'_2 + (1-p'_2)u(y_1)}{\delta} < u(y_2 - \bar{v}) + \frac{u(y_2) + u(y_1)}{2\delta}$ i.e.

$$\frac{(1/2 - p'_2)(u(y_2) - u(y_1))}{\delta} > u(y_2) - u(y_2 - \bar{v})$$

The inferences on how $y_2 - y_1$, $\bar{v}$ and δ influence the participation (see Section 3.2.3) remain the same for the above condition. To discuss the more general Nash equilibria, consider the conditions for all of the Cases I, II, III and IV to become a Nash equilibrium. For ease of notation we define the following for a given $\bar{v}$, y_1 and y_2

$$U_1 \equiv u(y_1) - u(y_1 - \bar{v})$$

$$U_2 \equiv u(y_2) - u(y_2 - \bar{v})$$

One could view U_1 and U_2 as the immediate gain in wealth utility from withdrawing. If $u(\cdot)$ represents a diminishing utility, then it is also necessary for $y_1 < y_2$ that

$$U_2 < U_1$$

Consider now that conditions for Case I to be a Nash Equilibrium. If Case I were to be a Nash equilibrium, then for the rich participant's decision to participate, the poor participant must get more utility from withdrawing than from participating and for poor participant's decision to withdraw, the rich must get more utility from participating than from withdrawing. The following conditions³ must therefore hold simultaneously

$$\begin{aligned} \frac{(\frac{1}{N} - p_1)(u(y_2) - u(y_1))}{\delta} &< U_1 \\ U_2 &< \frac{(p_2 - 1/N)(u(y_2) - u(y_1))}{\delta} \end{aligned} \quad (7)$$

Since $p_2 - \frac{1}{N} \geq \frac{1}{N} - p_1$ (strict inequality follows when the number of poor participants P is higher than that for rich participants $N - P$ e.g., with $N = 3$ and $P = 2$), this is a fairly general condition for the equilibrium. An increase in U_1 due to higher $\bar{v}$ does not change the equilibrium conditions so long as the corresponding rise U_2 (due to higher $\bar{v}$) does not affect the second inequality. In other words, a higher $\bar{v}$ favours the equilibrium (Case I) so long as there is an incentive for the rich to participate. Likewise, a decrease in $\bar{v}$ would be favourable for the equilibrium (Case I) so long as the poor participants do not also have an incentive to participate.

Similar effects can be seen with income differences $y_2 - y_1$ (which increase $u(y_2) - u(y_1)$ as well as $\frac{U_2}{U_1}$) if we rewrite Equation 7 as

$$\begin{aligned} \frac{(\frac{1}{N} - p_1)}{\delta} &< \frac{U_1}{u(y_2) - u(y_1)} \\ \frac{U_2}{(u(y_2) - u(y_1))} &< \frac{(p_2 - 1/N)}{\delta} \end{aligned}$$

³ $u(y_1) + \frac{p_1 u(y_2) + (1-p_1)u(y_1)}{\delta} > u(y_1 - \bar{v}) + \frac{\frac{1}{N}u(y_2) + \frac{1}{N}u(y_1)}{\delta} \Rightarrow u(y_1) - u(y_1 - \bar{v}) > \frac{(\frac{1}{N} - p_1)(u(y_2) - u(y_1))}{\delta}$ and $u(y_2 - \bar{v}) + \frac{p_2 u(y_2) + (1-p_2)u(y_1)}{\delta} > u(y_2) + \frac{\frac{1}{N}u(y_2) + \frac{1}{N}u(y_1)}{\delta} \Rightarrow u(y_2) - u(y_2 - \bar{v}) < \frac{(p_2 - 1/N)(u(y_2) - u(y_1))}{\delta}$

Limiting our attention to cases when $\bar{v} < y_1$, higher y_2 for fixed y_1 decreases $\frac{U_2}{u(y_2)-u(y_1)}$ and makes it more likely for the rich to participate while it also decreases $\frac{U_1}{u(y_2)-u(y_1)}$ by raising the stakes enough for the poor to participate (making it less likely for first inequality to be fulfilled). It is easy to see that higher impatience δ encourages both poor and rich to withdraw. As would be evident, the equilibrium conditions corresponding to Case II are far more stringent.

If Case II is a Nash Equilibrium, then for rich's decision to withdraw, the poor participant should get more utility from participating than from withdrawing and for poor participant's decision to participate, the rich participant should get more utility from withdrawing than from participating. This means that the following hold ⁴ simultaneously,

$$\begin{aligned} U_1 &< \frac{(p'_1 - 1/N)(u(y_2) - u(y_1))}{\delta} \\ U_2 &> \frac{(1/N - p'_2)(u(y_2) - u(y_1))}{\delta} \end{aligned}$$

Since $\frac{(p'_1 - 1/N)(u(y_2) - u(y_1))}{\delta} < \frac{(1/N - p'_2)(u(y_2) - u(y_1))}{\delta}$, the above cannot be satisfied so long as we have a diminishing utility which necessitates $U_2 < U_1$. Thus an equilibrium with Case II which incentivises the poor to participate but not the rich is possible only with a reference dependent utility with $U_1 < U_2$. With $U_1 < U_2$, we can write above as

$$\frac{U_1}{u(y_2) - u(y_1)} < \frac{(p'_1 - 1/N)}{\delta} < \frac{(1/N - p'_2)}{\delta} < \frac{U_2}{u(y_2) - u(y_1)}$$

Notice that despite a reference-dependent utility, a higher $\bar{v}$ or more impatience (δ) may disincentivise the poor from participation just the same (leading to Case III). Similarly, higher income differences (an increase in y_2 for fixed y_1) could encourage the rich to participate by decreasing $\frac{U_2}{u(y_2)-u(y_1)}$ (leading to Case IV). Even with the reference dependent utility, the equilibrium with Case II remains a fragile one.

If Case III were to be a Nash-Equilibrium, then for rich participant's decision to withdraw, poor should get more utility from withdrawing than from participating and for poor participant's decision to withdraw, the rich participant should get more from withdrawing than from participating. The following should thus hold⁵,

$$\begin{aligned} U_1 &> \frac{(p'_1 - \frac{1}{N})(u(y_2) - u(y_1))}{\delta} \\ U_2 &> \frac{(p_2 - \frac{1}{N})(u(y_2) - u(y_1))}{\delta} \end{aligned}$$

Since $p'_1 - \frac{1}{N} \leq p_2 - \frac{1}{N}$ (strict inequality when number of poor participants is higher), we have $\frac{(p'_1 - \frac{1}{N})(u(y_2) - u(y_1))}{\delta} < \frac{(p_2 - \frac{1}{N})(u(y_2) - u(y_1))}{\delta}$ and $U_2 < U_1$. A higher $\bar{v}$ pushes both U_2 and U_1 further away from participation but a fall in $\bar{v}$, incentivises the rich participant to participate (leading to Case I).

$$\begin{aligned} \frac{U_1}{u(y_2) - u(y_1)} &> \frac{(p'_1 - \frac{1}{N})}{\delta} \\ \frac{U_2}{u(y_2) - u(y_1)} &> \frac{(p_2 - \frac{1}{N})}{\delta} \end{aligned}$$

⁴ $u(y_1 - \bar{v}) + \frac{p'_1 u(y_2) + (1-p'_1)u(y_1)}{\delta} > u(y_1) + \frac{\frac{1}{N}u(y_2) + \frac{1}{N}u(y_1)}{\delta} \Rightarrow u(y_1) - u(y_1 - \bar{v}) < \frac{(p'_1 - 1/N)(u(y_2) - u(y_1))}{\delta}$ and $u(y_2) + \frac{p'_2 u(y_2) + (1-p'_2)u(y_1)}{\delta} > u(y_2 - \bar{v}) + \frac{\frac{1}{N}u(y_2) + \frac{1}{N}u(y_1)}{\delta} \Rightarrow u(y_2) - u(y_2 - \bar{v}) > \frac{(1/N - p'_2)(u(y_2) - u(y_1))}{\delta}$
⁵ $u(y_1) + \frac{\frac{1}{N}u(y_2) + \frac{1}{N}u(y_1)}{\delta} > u(y_1 - \bar{v}) + \frac{p'_1 u(y_2) + (1-p'_1)u(y_1)}{\delta} \Rightarrow u(y_1) - u(y_1 - \bar{v}) > \frac{(p'_1 - \frac{1}{N})(u(y_2) - u(y_1))}{\delta}$ and $u(y_2) + \frac{\frac{1}{N}u(y_2) + \frac{1}{N}u(y_1)}{\delta} > u(y_2 - \bar{v}) + \frac{p_2 u(y_2) + (1-p_2)u(y_1)}{\delta} \Rightarrow u(y_2) - u(y_2 - \bar{v}) > \frac{(p_2 - \frac{1}{N})(u(y_2) - u(y_1))}{\delta}$

Higher income differences i.e. higher y_2 for fixed y_1 also decrease $\frac{U_2}{u(y_2)-u(y_1)}$ and incentivise the rich participant to participate (leading to an equilibrium with Case I). Conversely, if the game is not lucrative for the rich it would automatically be infeasible for the poor i.e. if rich withdraws then poor automatically withdraws. If there is any net utility from playing the game, however, Case IV is the likely equilibrium.

If Case IV were to be a Nash Equilibrium, then for rich's decision to participate, the poor participant should get more from participating than from withdrawing and for poor's decision to participate, the rich participant should get more from participating than from withdrawing. Therefore, the following⁶ must hold

$$U_1 < \frac{(\frac{1}{N} - p_1)(u(y_2) - u(y_1))}{\delta}$$

$$U_2 < \frac{(\frac{1}{N} - p'_2)(u(y_2) - u(y_1))}{\delta}$$

With $\frac{(\frac{1}{N} - p'_2)(u(y_2) - u(y_1))}{\delta} < \frac{(\frac{1}{N} - p_1)(u(y_2) - u(y_1))}{\delta}$ and $U_2 < U_1$, this equilibrium rests upon the incentive for the poor participant to participate. So long as the poorer participant finds the competition worthwhile, the rich would also participate.

The above results outline how participation costs influence the investments towards status. For very low $\bar{v}$, rich and poor would both participate and for very high $\bar{v}$, both would withdraw. For everything in between, the only equilibrium may be the one where only the rich participate. A reference-based utility may permit an equilibrium but at the likely cost of segregation of income classes.

4 Conclusion

The paper examines how inequality can shape social capital investments—particularly education—when status is endogenised as both a determinant and an outcome of mobility. By considering the signaling value of education modeled as small-scale tournaments combining innate ability and social capital, the analysis links income differences, participation costs, and reference-dependent utility to long-term patterns of educational expenditure.

Three principal findings emerge. First, inequality can initially stimulate social capital investments, but excessive inequality makes status competitions unsustainable unless participation costs adjust accordingly. Second, participation costs are crucial: when they remain high, only the rich participate in status-related investments such as education, while lower costs allow both rich and poor to compete. Status competitions imply that a rise in inequality increases social capital expenditures when participation costs are held constant. The presence of signaling thus necessitates that participation costs rise to sustain limits to mobility. Status consideration, participation costs and inequality remain intricately linked so long as status competitions exist in a society. Reducing income-differences without lowering of participation costs in status competitions may have an effect of supporting status-related competitions under long-established inequalities. Third, under diminishing utility of wealth, the model admits only two stable long-run equilibria—either universal participation in status competitions or participation limited to the rich. These outcomes quantitatively formalize earlier qualitative arguments about the social limits to growth and mobility (Hirsch, 1977; Galbraith, 1958, 1994).

The inclusion of reference-dependent utility further highlights how perceived contentment can sustain segregation of competitions even under large income gaps—consistent with empirical evidence on meritocratic beliefs amid inequality (Mijs & Savage, 2020). Together, these results reveal a dynamic interplay between inequality, participation costs, and status incentives. For education

⁶ $u(y_1 - \bar{v}) + \frac{\frac{1}{N}u(y_2) + \frac{1}{N}u(y_1)}{\delta} > u(y_1) + \frac{p_1u(y_2) + (1-p_1)u(y_1)}{\delta} \Rightarrow u(y_1) - u(y_1 - \bar{v}) < \frac{(\frac{1}{N} - p_1)(u(y_2) - u(y_1))}{\delta}$ and $u(y_2 - \bar{v}) + \frac{\frac{1}{N}u(y_2) + \frac{1}{N}u(y_1)}{\delta} > u(y_2) + \frac{p'_2u(y_2) + (1-p'_2)u(y_1)}{\delta} \Rightarrow u(y_2) - u(y_2 - \bar{v}) < \frac{(\frac{1}{N} - p'_2)(u(y_2) - u(y_1))}{\delta}$

policy, they suggest that mobility cannot be expanded merely by widening access or increasing expenditure; the signaling role of education and the costs of participation must also be managed to prevent inequality from reproducing itself through status-driven competition.

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Appendices

A Convolution of Uniform Random Variables

The distribution of a convex combination $w = \mu r + (1 - \mu)s$ of two random variables r, s - which are distributed according to densities $f_R(r)$ and $f_S(s)$ - can be derived using the convolution of two derived variables $x = \mu r$ and $y = (1 - \mu)s$. If r and s are uniformly distributed variables ranging from 0 to 1, then x ranges from 0 to μ and y ranges from 0 to $(1 - \mu)$. The CDF for $F_X(x)$ and $F_Y(y)$ are simply $F_X(x) = \frac{x}{\mu}$, $F_Y(y) = \frac{y}{1-\mu}$ (respectively). Therefore, the densities for x and y are $f_X(x) = \frac{1}{\mu}$ and $f_Y(y) = \frac{1}{1-\mu}$. The density of w is simple $f(w) = \int_{-\infty}^{\infty} f_Y(w - x)f_X(x)dx$. Since, $f_X(x)$ is $1/\mu$ only between 0 and μ . We have the following integral to evaluate for the convolution

$$f(w) = \frac{1}{\mu} \int_0^{\mu} f_Y(w - x)dx$$

Since $f_Y(y)$ is 1 only between 0 and $1 - \mu$, we have the condition $0 < w - x < 1 - \mu \Rightarrow w - (1 - \mu) < x < w$. Further, x is distributed only between 0 and μ . As shown in Figure 2, the integral would evaluate to the following⁷ under the assumption $\mu < \frac{1}{2}$.

$$f(w) = \frac{1}{\mu} \int_0^{\mu} f_Y(w - x)dx = \begin{cases} \frac{1}{\mu} \int_0^w \frac{1}{1-\mu} dx & 0 < w < \mu \\ \frac{1}{\mu} \int_0^{\mu} \frac{1}{1-\mu} dx & \mu < w < 1 - \mu \\ \frac{1}{\mu} \int_{w-(1-\mu)}^{\mu} \frac{1}{1-\mu} dx & 1 - \mu < w < 1 \\ 0 & \text{otherwise} \end{cases}$$

This can be simplified as

$$f(w) = \begin{cases} \frac{w}{\mu(1-\mu)} & 0 < w < \mu \\ \frac{1}{1-\mu} & \mu < w < 1 - \mu \\ \frac{1-w}{\mu(1-\mu)} & 1 - \mu < w < 1 \\ 0 & \text{otherwise} \end{cases}$$

Figure 3 shows this trapezoidal density with μ set to $\frac{1}{5}$. The CDF $F_X(x)$ can be evaluated by integrating the above density as $P(X < x) = \int_{-\infty}^x f(t)dt$.

⁷Notice that as a convex combination, w is distributed only between 0 and 1.

$$F_X(x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{\mu}{2(1-\mu)} + \frac{x-\mu}{1-\mu} & \mu < x < 1-\mu \\ \frac{\mu^2}{2\mu(1-\mu)} + \frac{2\mu(1-2\mu)}{2\mu(1-\mu)} + \frac{\mu^2-x^2+2x-1}{2\mu(1-\mu)} & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Simplifying above, we have

$$F_X(x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{x-\mu/2}{1-\mu} & \mu < x < 1-\mu \\ 1 - \frac{(x-1)^2}{2\mu(1-\mu)} & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

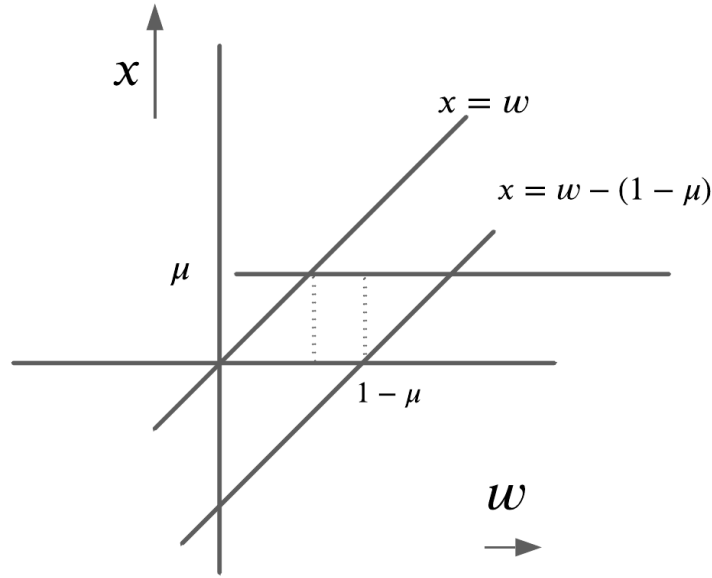


Figure 2: Ranges of w and $w - x$ for convolution of two scaled random variables

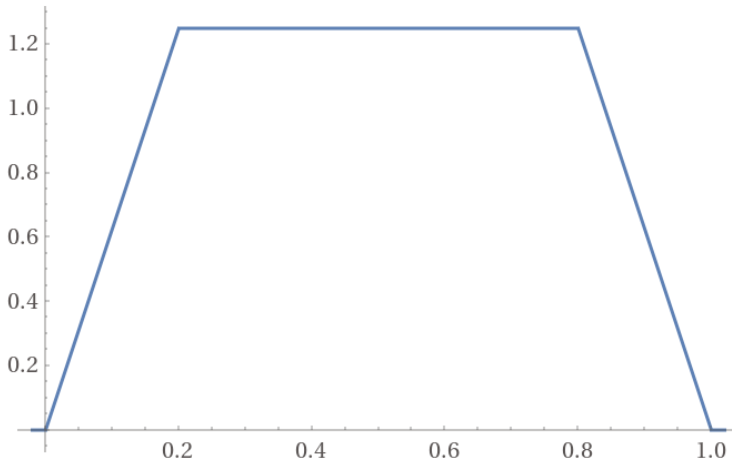


Figure 3: Combined density of w as the convex combination of two random variables with $\mu = \frac{1}{5}$

B $P(\mathbf{win}|r, s)$

To calculate $\int_{s=0}^{s=1} \int_{r=0}^{r=1} P(\mathbf{win}|r, s) dr ds$ over the variable x (where $0 < r < 1$ and $0 < s < 1$), we evaluate the sum of three parts split by the boundaries of the piecewise function. The three parts are shown in Figure 4. If $x = \mu r + (1 - \mu)s$ and $F(x)$ is of the following piece-wise form

$$F(x) = \begin{cases} A & 0 < x < \mu \\ B & \mu < x < 1 - \mu \\ C & 1 - \mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Then the integral $\int_{s=0}^{s=1} \int_{r=0}^{r=1} F(x) dr ds$ can be written as

$$\int_{s=0}^{s=1} \int_{r=0}^{r=1} F(x) ds dr = \int_{r=0}^{r=1} \int_{s=0}^{s=\frac{\mu(1-r)}{1-\mu}} A ds dr + \int_{r=0}^{r=1} \int_{s=\frac{\mu(1-r)}{1-\mu}}^{s=1-\frac{\mu r}{1-\mu}} B ds dr + \int_{r=0}^{r=1} \int_{s=1-\frac{\mu r}{1-\mu}}^{s=1} C ds dr$$

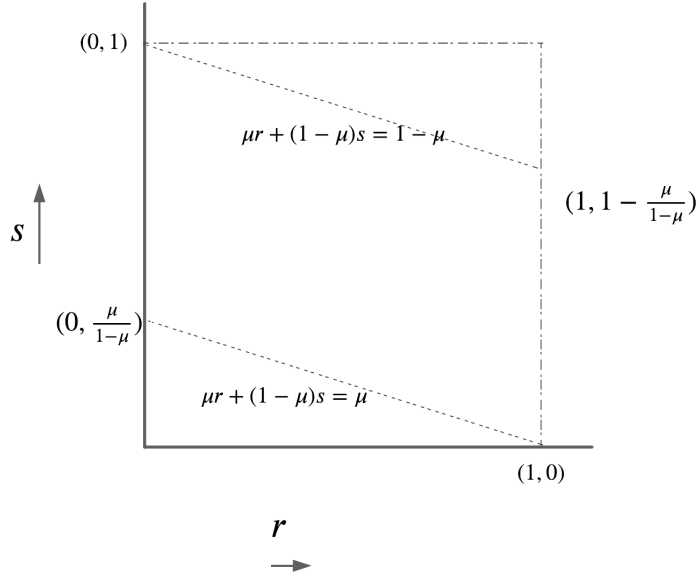


Figure 4: Integration over r, s split by boundaries $\mu r + (1 - \mu)s = \mu$ and $\mu r + (1 - \mu)s = 1 - \mu$